Nathanael Bosch

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Research Interests

I develop machine learning methods to make scientific simulation faster, more accurate, and more reliable. My current focus is on neural operators for electronic-structure simulation, in particular on learning the exchange-correlation functional in density functional theory. Before that, I worked on filtering-based probabilistic ODE solvers, making them more competitive with classical solvers and using them for robust parameter inference in mechanistic models. In all my work, I take inspiration from established numerical algorithms to build methods that actually work in practice, and I write open-source software to make my research reproducible and accessible.

Keywords: Scientific machine learning, neural operators, uncertainty quantification, probabilistic numerics, Gaussian processes, Bayesian filtering and smoothing, differential equations, parameter inference, density functional theory.

Professional Experience

Postdoctoral Researcher, EPFL, Lausanne, Switzerland 2025/10 – today
Machine learning for electronic-structure calculations, in the group of Michael Herbst (Mathematics for Materials Modelling). Developing SE(3)-equivariant Fourier neural operators for variable-cell crystal structures, with applications to learning exchange-correlation functionals for density functional theory (DFT), orbital-free DFT, and electron density prediction.

Applied Scientist Intern, Amazon Web Services, Berlin, Germany 2024/08 – 2025/01
Multi-layer stack ensembling for time series forecasting, in the Machine Learning and Forecasting team of AWS AI Research. The method is now part of the AutoGluon software package and resulted in an AutoML 2025 paper.

Doctoral Researcher, University of Tübingen, Germany 2020/05 – 2024/06
Probabilistic numerical methods for differential equations, in the group of Philipp Hennig (Methods of Machine Learning). Includes a two-month research stay with Simo Särkkä at Aalto University in spring 2022, which resulted in a JMLR paper.

Student Assistant, Max Planck Institute for Intelligent Systems, Tübingen, Germany 2019/03 – 2019/10
Master's thesis research in the group of Jörg Stückler (Embodied Vision). Resulted in an L4DC 2020 paper.

Education

Ph.D. Computer Science, University of Tübingen, Germany 2020/05 – 2025/02
Magna Cum Laude (with great distinction).
Thesis: "A Flexible and Efficient Framework for Probabilistic Numerical Simulation and Inference". Advisor: Philipp Hennig. Part of the International Max Planck Research School for Intelligent Systems (IMPRS-IS).

M.Sc. Data Engineering and Analytics, Technical University of Munich, Germany 2018/04 – 2019/10
Grade 1.3 (Distinction). Thesis: "Learning Gaussian Process Dynamics Models from Visual Observations for Control".

M.Sc. Mathematics, Technical University of Munich, Germany 2016/10 – 2018/10
Grade 1.2 (High Distinction). Thesis: "Evolutionary Games for Global Function Minimization".

B.Sc. Mathematics, Technical University of Munich, Germany 2012/10 – 2016/06
Grade 1.8 (Distinction). Thesis: "Different Noise Models in Variable Density Compressed Sensing".

Awards, Grants & Scholarships

Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowship 2026
Proposal "Gradient-Accelerated and Error-Aware Multi-Fidelity Bayesian Optimization for Inverse Materials Design", score 96.2% (funding threshold 96.8%).

Travel Grant, Magnus Ehrnrooth Foundation 2026
Travel funding for the 2nd International Conference on Probabilistic Numerics (ProbNum26).

Outstanding Reviewer Awards 2022 – 2026
ICML 2022, ICML 2023, NeurIPS 2026.

Leadership Talent Academy, Karl Schlecht Stiftung 2025
Scholarship (20 per cohort) for a 3-month leadership programme at the University of Tübingen, with three 3-day modules and individual coaching.

Research Visits Abroad Grant, Cluster of Excellence Machine Learning, University of Tübingen 2022
Funding for the research visit to Aalto University.

Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation) 2013 – 2019
Merit-based scholarship for undergraduate and graduate studies.

Publications

Selected publications

Nathanael Bosch. *"Stable and Scalable Probabilistic Numerical Solvers for Stiff and High-Dimensional ODEs"*. Conference on Neural Information Processing Systems, **NeurIPS**, 2026.

Nathanael Bosch*, Niklas Frederik Schmitz*, and Michael F. Herbst. *"Euclidean Fourier Neural Operators"*. Representations for the Physical Sciences Workshop at NeurIPS, 2026.

Jonas Beck, Nathanael Bosch, Michael Deistler, Kyra L. Kadhim, Jakob H. Macke, Philipp Hennig, and Philipp Berens. *"Diffusion Tempering Improves Parameter Estimation with Probabilistic Integrators for Ordinary Differential Equations"*. International Conference on Machine Learning, **ICML**, 2024.

Nathanael Bosch, Adrien Corenflos, Fatemeh Yaghoobi, Filip Tronarp, Philipp Hennig, and Simo Särkkä. *"Parallel-in-Time Probabilistic Numerical ODE Solvers"*. Journal of Machine Learning Research, **JMLR**, 2024.

Nathanael Bosch, Philipp Hennig, and Filip Tronarp. *"Probabilistic Exponential Integrators"*. Conference on Neural Information Processing Systems, **NeurIPS**, 2023.

Further publications

Joanna Sliwa, Frank Schneider, Nathanael Bosch, Agustinus Kristiadi, and Philipp Hennig. *"Low-Rank Filtering & Smoothing for Sequential Deep Learning"*. Transactions on Machine Learning Research, **TMLR**, 2026.

Nathanael Bosch, Oleksandr Shchur, Nick Erickson, Michael Bohlke-Schneider, and Ali Caner Turkmen. *"Multi-layer Stack Ensembles for Time Series Forecasting"*. International Conference on Automated Machine Learning, **AutoML**, 2025.

Ori Press, Brandon Amos, Haoyu Zhao, Yikai Wu, Samuel Ainsworth, Dominik Krupke, Patrick Kidger, Touqir Sajed, Bartolomeo Stellato, Jisun Park, Nathanael Bosch, Eli Meril, Albert Steppi, Arman Zharmagambetov, Fangzhao Zhang, David Pérez-Piñeiro, Alberto Mercurio, Ni Zhan, Talor Abramovich, Kilian Lieret, Hanlin Zhang, Shirley Huang, Matthias Bethge, and Ofir Press. *"AlgoTune: Can Language Models Speed Up General-Purpose Numerical Programs?"* Conference on Neural Information Processing Systems, **NeurIPS** (Datasets and Benchmarks Track), 2025.

Dingling Yao, Filip Tronarp, and Nathanael Bosch. *"Propagating Model Uncertainty through Filtering-based Probabilistic Numerical ODE Solvers"*. International Conference on Probabilistic Numerics, **ProbNum**, 2025.

Nathanael Bosch. *"ProbNumDiffEq.jl: Probabilistic Numerical Solvers for Ordinary Differential Equations in Julia"*. Journal of Open Source Software, **JOSS**, 2024.

Amon Lahr, Filip Tronarp, Nathanael Bosch, Jonathan Schmidt, Philipp Hennig, and Melanie N. Zeilinger. *"Probabilistic ODE Solvers for Integration Error-aware Numerical Optimal Control"*. Learning for Dynamics & Control Conference, **L4DC**, 2024.

Nathanael Bosch, Filip Tronarp, and Philipp Hennig. *"Pick-and-Mix Information Operators for Probabilistic ODE Solvers"*. International Conference on Artificial Intelligence and Statistics, **AISTATS**, 2022.

Nicholas Krämer*, Nathanael Bosch*, Jonathan Schmidt*, and Philipp Hennig. *"Probabilistic ODE Solutions in Millions of Dimensions"*. International Conference on Machine Learning, **ICML**, 2022.

Filip Tronarp*, Nathanael Bosch*, and Philipp Hennig. *"Fenrir: Physics-Enhanced Regression for Initial Value Problems"*. International Conference on Machine Learning, **ICML**, 2022.

Nathanael Bosch, Philipp Hennig, and Filip Tronarp. *"Calibrated Adaptive Probabilistic ODE Solvers"*. International Conference on Artificial Intelligence and Statistics, **AISTATS**, 2021.

Jonathan Wenger, Nicholas Krämer, Marvin Pförtner, Jonathan Schmidt, Nathanael Bosch, Nina Effenberger, Johannes Zenn, Alexandra Gessner, Toni Karvonen, François-Xavier Briol, Maren Mahsereci, and Philipp Hennig. *"ProbNum: Probabilistic Numerics in Python"*. arXiv, 2021.

Nathanael Bosch*, Jan Achterhold*, Laura Leal-Taixé, and Jörg Stückler. *"Planning from Images with Deep Latent Gaussian Process Dynamics"*. Learning for Dynamics & Control Conference, **L4DC**, 2020.

* indicates equal contribution.

Selected Software

ProbNumDiffEq.jl, maintainer, Julia 2020 – today
Probabilistic numerical ODE solvers based on Bayesian filtering and smoothing. Integrated with DifferentialEquations.jl and part of the downstream tests of OrdinaryDiffEq.jl. Published in JOSS (2024). 130+ GitHub stars.

probnum, core contributor, Python 2020 – 2021
Python library for probabilistic numerics, including ODE solvers, Bayesian state estimation, linear solvers, and quadrature. 460+ GitHub stars.

Further contributions 2021 – 2026
SciML ecosystem (OrdinaryDiffEq.jl, SciMLBase.jl, LinearSolve.jl), DFTK.jl, AutoGluon.

Talks

Invited talks & tutorials

Probabilistic Numerics Workshop (ProbNum24), London, United Kingdom 2024/07
Talk: "Robust Parameter Inference in ODEs via Physics-Enhanced Gaussian Process Regression"

Probabilistic Numerics Spring School, Southampton, United Kingdom 2024/04
Tutorial: "Probabilistic Numerics for ODEs"

International Conference on Scientific Computing and Machine Learning, Kyoto, Japan 2024/03
Talk: "Probabilistic Numerical Solvers for Differential Equations"

SIAM Conference on Uncertainty Quantification, Trieste, Italy 2024/02
Talk: "Probabilistic Numerics for Ordinary Differential Equations"

Statistics PhD Seminar, University of Edinburgh, Edinburgh, United Kingdom (virtual) 2023/05
Talk: "Probabilistic Numerics for Ordinary Differential Equations"

Probabilistic Numerics Spring School, Tübingen, Germany 2023/03
Tutorial: "Probabilistic Numerics for ODEs"

Institute for Dynamic Systems and Control, ETH Zurich, Zurich, Switzerland 2023/01
Talk: "ODE Solvers as Gauss–Markov Regression"

Sensor Informatics and Medical Technology Group, Aalto University, Espoo, Finland 2022/05
Talk: "Solving Differential Equations with Bayesian Filtering and Smoothing"

Dagstuhl Seminar on Probabilistic Numerical Methods, Wadern, Germany 2021/10
Talk: "ProbNumDiffEq.jl: Fast and Practical ODE Filters in Julia"

Contributed talks

International Conference on Probabilistic Numerics (ProbNum25), Antibes, France 2025/09
Talk: "Propagating Model Uncertainty through Filtering-based Probabilistic Numerical ODE Solvers"

JuliaCon 2024, Eindhoven, Netherlands 2024/07
Talk: "Fast Probabilistic Inference for ODEs with ProbNumDiffEq.jl"

Teaching

Lectures & courses

Lecture in Probabilistic Machine Learning (M.Sc.), University of Tübingen 2023
"Hidden Markov Models", in Philipp Hennig's course.

Two lectures in Numerics of Machine Learning (M.Sc.), University of Tübingen 2022 / 2023
"Ordinary Differential Equations" and "Probabilistic Numerical ODE Solvers", in a course taught jointly by Philipp Hennig and members of his group. Also prepared and graded exercise sheets.

Guest lecture at Uppsala University 2022
"Probabilistic Numerics for Ordinary Differential Equations", in Zheng Zhao's seminar "A computational introduction to stochastic differential equations".

Seminar "Machine Learning for and with Dynamical Systems" (M.Sc.), University of Tübingen 2021
Organised jointly with Nicholas Krämer and Philipp Hennig.

Abitur preparation course at abiturma GbR, Munich and Stuttgart 2016, 2017
Sole instructor for a class of final-year school students in a five-day intensive course in mathematics.

Teaching assistance

Analysis I (B.Sc.), EPFL 2026 / 2027
Exercise sessions in French, helping students with the problem sets. Course by Nicolas Boumal.

Numerical Analysis (B.Sc.), EPFL 2026
Exam grading, for a course taken by materials science, electrical engineering, and chemistry students.

Linear Algebra I (B.Sc.), EPFL 2025 / 2026
Exercise sessions in French, helping students with the problem sets. Graded exercise sheets and exams. Course by Philippe Michel.

Probabilistic Machine Learning (M.Sc.), University of Tübingen 2023
Prepared and presented individual exercise sessions and helped grade homework and exams. Course by Philipp Hennig.

Data Literacy (M.Sc.), University of Tübingen 2021 / 2022
Supervised student projects, including feedback and grading. Also prepared and presented individual exercise sessions and helped grade homework and exams. Course by Philipp Hennig.

Time Series (M.Sc.), University of Tübingen 2020 / 2021
Exercise sessions every second week throughout the semester, preparing and presenting the solutions. Graded homework and exams. Course by Filip Tronarp.

Principles of Mathematics 2 (B.Sc.), Technical University of Munich 2017
Weekly exercise session for Engineering Science students, preparing and presenting the solutions.

Supervision

Miguel Francisco Magro Dos Reis, Master's semester project 2026
"Euclidean Fourier Neural Operators for Electronic Density Prediction".

Andrés Santiago Ulloa Canizares, Bachelor's student project 2026
"Bayesian Optimization for Crystal Structure Prediction".

Fabrice van der Lehr, Master's thesis 2024
"Bayesian Filtering for Black Box Simulators". Co-supervised with Tobias Weber.

Sofiya Garkot, Essay rotation 2023 / 2024
"Exploring Probabilistic Solvers for Hodgkin-Huxley Models". Co-supervised with Jonas Beck.

Thomas Albrecht, Master's thesis 2021 / 2022
"Bayesian Physics-Informed Neural Networks via Laplace Approximations".

Felix Böhm, Bachelor's thesis 2021 / 2022
"Inferring ODE Parametric Latent Forces via Neural ODEs".

Dingling Yao, Master's thesis 2021
"Uncertainty Propagation in Probabilistic Ordinary Differential Equation Solvers". Co-supervised with Filip Tronarp. Resulted in a ProbNum 2025 paper.

Joanna Sliwa, Essay rotation 2021
"Physics-Informed Neural Networks". Co-supervised with Nicholas Krämer.

Didactic training

Effective Teaching & Mentoring: An Immersive Programme, EPFL 2026
60-hour programme on evidence-based course design, classroom teaching, project-based learning, assessment, and mentoring in higher education. In-person workshops, independent study, and a final report.

Academic Services

Program committee, 2nd International Conference on Probabilistic Numerics (ProbNum26) 2026

Review services

International Conference on Machine Learning (ICML)	2022 – 2024
Conference on Neural Information Processing Systems (NeurIPS)	2021 – 2023, 2026
International Conference on Artificial Intelligence and Statistics (AISTATS)	2021 – 2024, 2026 – 2027
Transactions on Machine Learning Research (TMLR)	2022 – 2023
International Conference on Probabilistic Numerics (ProbNum)	2025 – 2026

Language Skills

Languages: German and French (native), English (fluent), Spanish (intermediate), Russian (basic).

Teaching: German, English, French.

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